

Asmaa Abdul-Amin

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Summary

Software Engineer with 2+ years of experience building production software, REST APIs, backend systems, and automation tools supporting NASA mission platforms at The Johns Hopkins University Applied Physics Laboratory. Experienced across the full software development lifecycle using Python, PHP, JavaScript, SQL, and cloud-native development practices. **Previously held an Interim Secret clearance** (currently inactive and eligible for reactivation through a sponsoring employer).

Skills

- Programming Languages: Python, PHP, JavaScript, React, SQL, JSON, HTML, XML, ReqIF, CSS
 - Frameworks & Technologies: Atlassian, Bootstrap, Elasticsearch, IBM DOORS, REST APIs, Linux, Git
 - Libraries & Tools: Pandas, NumPy, SciPy, Torch, lxml, gymnasium, INotify, TensorFlow, Hugging Face, OpenAI API, Qiskit
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Experience

Johns Hopkins Applied Physics Laboratory

Software Engineering Intern

Space Exploration Sector (SOF-2) | May 2024 – May 2026

- Developed and maintained production web applications supporting 10+ active NASA and APL missions, improving content management and engineering workflows for mission scientists and technical teams.
- Engineered a Python automated ETL pipeline converting .reqIF/.XML files to .JSON for indexing into Elasticsearch powering a requirements intelligence pipeline that transforms static requirements documents into a searchable, machine-readable knowledge base.
- Built Python automated utilities and database export tools that generate JSON feeds from mission databases to support downstream scientific analysis and integration with APL internal engineering systems.
- Designed and implemented database-driven administrative portals and public-facing web applications using PHP, JavaScript, MySQL, Bootstrap, and DataTables, across more than a dozen NASA/APL mission platforms enabling mission teams to manage content without code changes.
- Modernized legacy NASA web applications by migrating archived mission data into APL databases and creating new interfaces with responsive components, improving usability and simplifying content discovery for mission stakeholders.
- Planned, estimated, and developed the JHUAPL Ice Giants mission website, including content migration, workshop archives, and administrative tooling.
- Collaborated directly with scientists, project managers, and technical staff to translate requirements into deployed software solutions supporting active NASA mission platforms.
- Owned features from requirements gathering and implementation through testing, deployment, maintenance, and iterative enhancements across multiple concurrent projects.

Undergraduate Research Intern

Research & Exploratory Development Department (R1N) | Mar 2023 – May 2024

- Deployed multiple webpages for the NASA Dragonfly internal portal, streamlining access to critical mission resources and improving mission collaboration efficiency.
 - Developed machine learning models analyzing COVID-19 policies vs. mortality, improving interpretability of public health data.
 - Presented research at NIST-PREP (Professional Research Experience Program) Showcase.
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Projects

Parker Solar Probe - Developed and maintained the Parker-Kit Celebration webpage.

Ice Giants - Planned, estimated, and implemented the Ice Giants research website.

EZIE - Rebuilt the EZIE website from ReactJS to PHP.

LOGIC - Implemented a database-backed terminology management system for standardized lunar infrastructure definitions.

LSIC - Created JSON APIs and optimized database-backed newsletter management.

BLKPVNTHR.OS – Engineering an AI-powered personal operating system featuring specialized autonomous agents and workflows.

Education

B.S. Data Science – University of Maryland Global Campus (Expected Dec 2028)

A.A. Computer Science – Montgomery College (Expected Dec 2026)

Certifications:

The Johns Hopkins University Applied Physics Laboratory – Python Programming II (2025)

Univ. of Chicago – Quantum Computing Systems Design I (2024)

IBM – Machine Learning with Python (2023)

Harvard – CS50 Intro to Computer Science (2022)